One-Way-ANOVA

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One Way ANOVA

Load data

```
data <- read.csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/One-Way-ANOVA/i
data$Fertilizer <- as.factor(data$Fertilizer) # convert to factor so ANOVA treats it as categ
```

Create ANOVA Model

To test whether mean yield differs among fertilizer groups

```
model <- aov(Yield ~ Fertilizer, data = data)
```

Assumption Checking

Shows one way anova is reliable or not.

1. Normality of Residuals

```r
shapiro.test(residuals(model))
```

```
    Shapiro-Wilk normality test

data:  residuals(model)
W = 0.82816, p-value = 0.01994
```

Interpretation:

- $p > 0.05 \rightarrow$ residuals are approximately normal $\rightarrow$ acceptable
- $p < 0.05 \rightarrow$ deviation from normality $\rightarrow$ check plots before deciding

Understanding Normality

- Normal data → values are evenly distributed around mean. Example: (15, 13, 14, 16)
- Non-normal data → extreme variation/outliers. Example: (2, 5, 20, 50)

2. Homogeneity of Variance

```r
library(car)
```

```
Warning: package 'car' was built under R version 4.4.3
```

```
Loading required package: carData
```

```r
leveneTest(Yield ~ Fertilizer, data = data)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2       0      1
       9
```

Interpretation

1. P-Value is more than 0.05, the variances across groups are more or less equal (homogeneous), perform the one-way ANOVA reliably.
2. P-Value is less than 0.05 the variances across groups are not more or less equal (not homogeneous), use caution. see the plot

Homogeneity

For understanding homogeneity need to understand variance. Variance means the spread of data in each group.

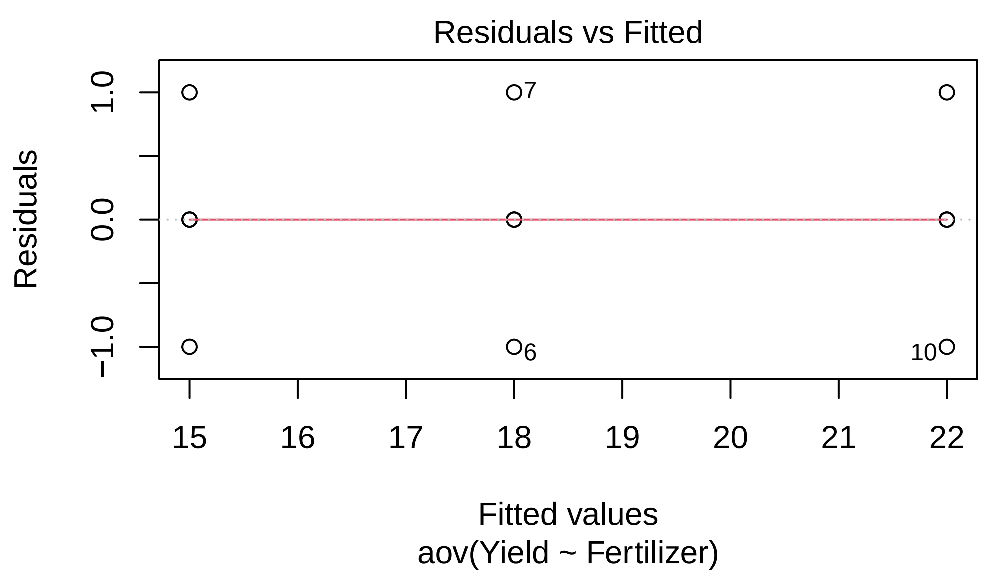
Example: Group A: (13, 14, 15, 16) → small spread → low variance

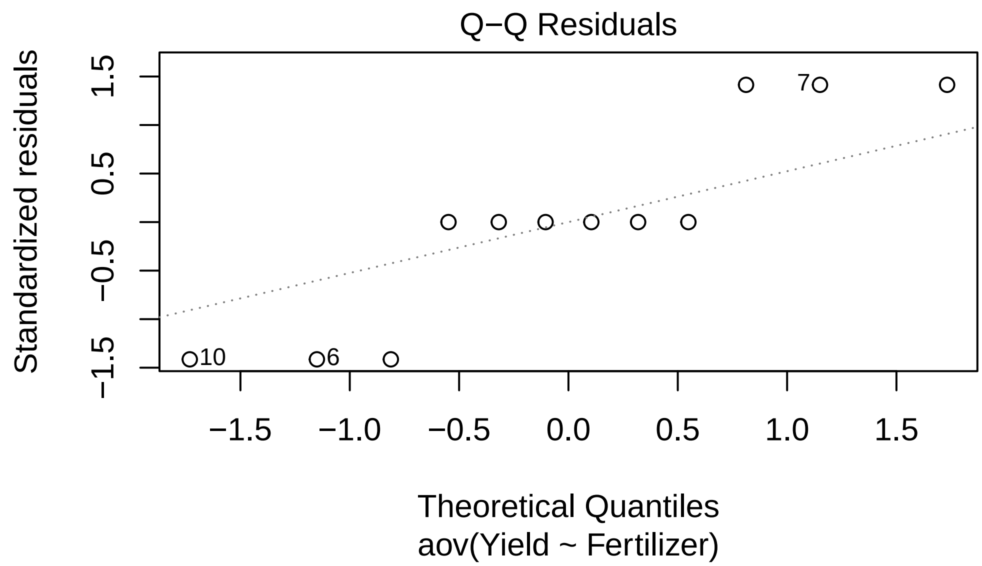
Group B: (10, 20, 30, 40) → large spread → high variance

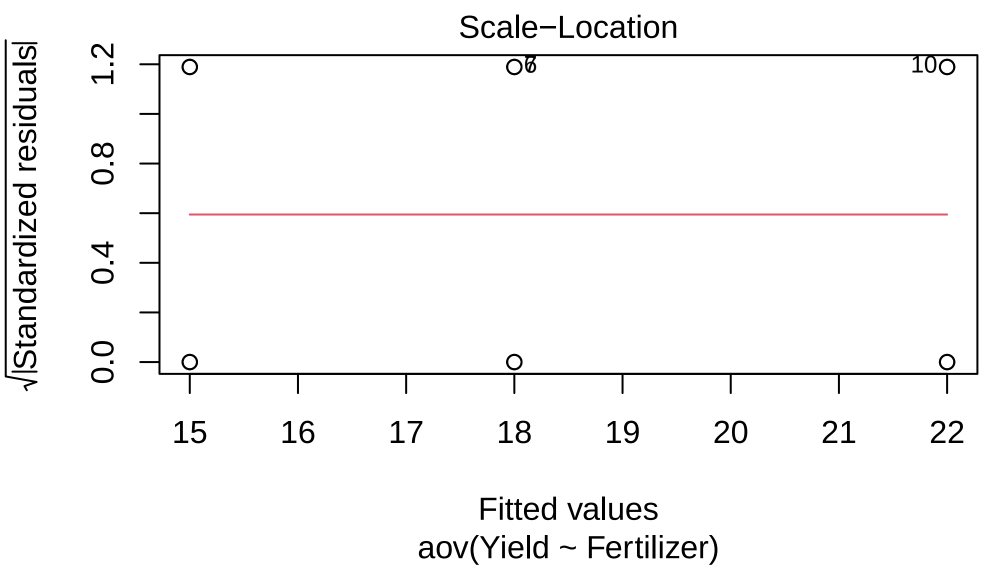
All groups have similar spread, the data is homogeneous. Ad one group is very spread out and others are tight, it's not homogeneous.

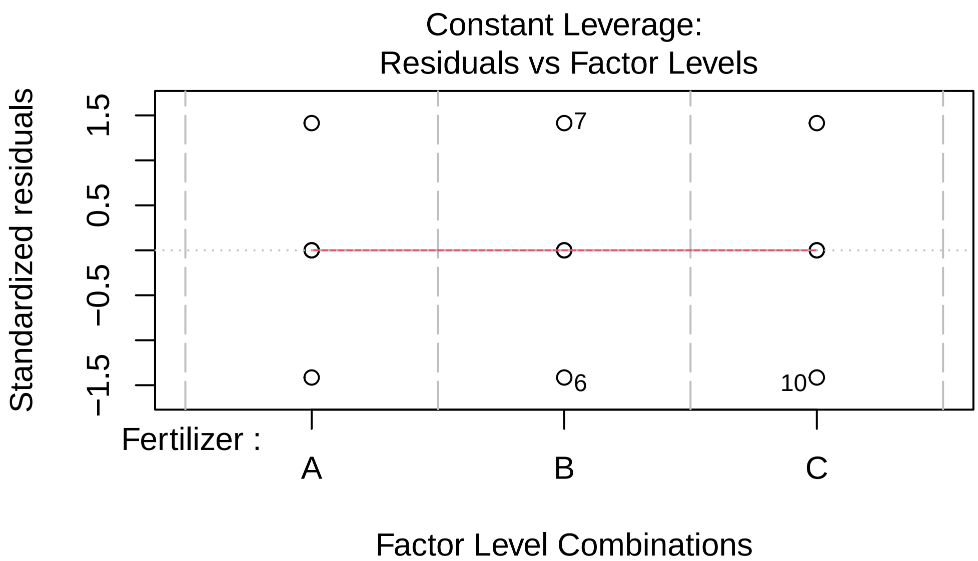
3. Diagnostic Plots

```r
plot(model)
```




Q−Q Residuals
Standardized residuals
1.5
0.5
−0.5
−1.5
10
6
7
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Theoretical Quantiles
aov(Yield ~ Fertilizer)


Scale−Location
√|Standardized residuals|
1.2
0.8
0.4
0.0
10
15
16
17
18
19
20
21
22
Fitted values
aov(Yield ~ Fertilizer)



What to check:

Residuals vs Fitted → should show random pattern (no structure)

Normal Q-Q → points should lie near the straight line

Scale-Location → spread should be consistent

Residuals vs Leverage → detect outliers/influential points

These plots are more reliable than tests. Study more here!!!!

Perform anova

```r
summary(model)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)    
Fertilizer   2  98.67   49.33      74 2.59e-06 ***
Residuals    9   6.00    0.67                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

$p < 0.05$ → significant difference exists among groups

$p > 0.05$ → no strong evidence of difference

ANOVA tells "difference exists". It does NOT tell which groups differ.

Alternative (Kruskal-Wallis Test)]

Use when

- strong non-normality
- outliers present
- variance highly unequal

```
krus <- read.csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/One-Way-ANOVA/r

kruskal.test(Yield ~ Fertilizer, data = krus)
```

```

	Kruskal-Wallis rank sum test

data:  Yield by Fertilizer
Kruskal-Wallis chi-squared = 4.3077, df = 2, p-value = 0.116
```

Interpretation:

$p < 0.05$ → at least one group differs

$p > 0.05$ → no evidence of difference

ANOVA clean workflow

```
# Load Data & Prepare
data <- read.csv("G:/Lesson/Done/Research/R Language/R-Language/Self-Learning/One-Way-ANOVA/a
data$Treatment <- as.factor(data$Treatment)  # ensure categorical variable

#Build ANOVA Model
model <- aov(Yield ~ Treatment, data = data)
```

```r
#Check Assumptions
##Normality of Residuals
shapiro.test(residuals(model))
```

```

	Shapiro-Wilk normality test

data:  residuals(model)
W = 0.93739, p-value = 0.3183
```

Interpretation

p > 0.05 → acceptable (not perfect, just no strong violation)

p < 0.05 → check plots before decision

```r
## Homogeneity of Variance

library(car)
leveneTest(Yield ~ Treatment, data = data)
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  3    0.75 0.5431
      12
```

Interpretation

p > 0.05 → equal variance (good)

p < 0.05 → violation (be careful)

Decision Rule (Important)

If assumptions roughly OK → proceed with ANOVA

If strong violation → go to non-parametric (Kruskal)

```r
#Run ANOVA
summary(model)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)    
Treatment    3  642.8  214.25   233.7 6.61e-11 ***
Residuals   12   11.0    0.92                     
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation

$p < 0.05$ → reject null → at least one group differs → go for post hoc

$p > 0.05$ → fail to reject → stop (no post hoc)